

BAKHTA, Russian Federation

WMO: 237760

Lat: 62.4629N

Lon: 89.0051E

Elev: 187

StdP: 14.60

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-47.2	-41.8	-50.6	0.3	-44.9	-46.3	0.4	-40.4	22.5	0.6	16.6	-2.6	0.7	320	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	83.0	65.5	79.8	64.1	76.5	63.1	68.4	77.4	66.5	75.2	64.8	72.9	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	94.5	72.3	63.4	87.9	70.2	61.6	82.5	68.9	32.9	76.3	31.4	75.6	30.0	72.7	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.3	13.4	DB	-51.7	88.6	3.6	2.4	-54.3	90.3	-56.3	91.7	-58.3	93.0	-60.9	94.7	(4)
(5)			WB	-49.6	73.1	1.9	2.9	-51.0	75.2	-52.1	76.9	-53.2	78.5	-54.6	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	-11.3	-3.6	15.4	28.7	39.6	60.4	65.6	57.9	44.6	27.8	2.4	-3.9	(6)
(7)		DBStd	28.74	17.60	15.32	13.32	9.55	8.77	9.07	6.54	6.78	7.74	9.68	15.68	18.23	(7)
(8)		HDD50	9481	1900	1500	1073	639	343	23	1	13	199	690	1429	1671	(8)
(9)		HDD65	13983	2365	1920	1538	1089	788	189	76	236	612	1155	1879	2136	(9)
(10)		CDD50	1134	0	0	0	0	21	334	484	258	37	0	0	0	(10)
(11)		CDD65	160	0	0	0	0	0	50	94	16	0	0	0	0	(11)
(12)		CDH74	1241	0	0	0	0	7	453	687	94	0	0	0	0	(12)
(13)		CDH80	247	0	0	0	0	1	90	146	10	0	0	0	0	(13)
(14)	Wind	WSAvg	5.2	4.5	4.6	5.2	5.9	6.2	5.0	4.6	4.8	5.3	6.6	5.0	4.4	(14)
(15)	Precipitation	PrecAvg	28.0	1.8	1.5	1.6	1.7	2.2	2.6	2.6	2.9	2.8	3.2	2.8	2.4	(15)
(16)		PrecMax	34.6	3.5	2.5	2.9	2.7	4.5	5.7	4.4	4.6	4.2	5.2	4.9	3.7	(16)
(17)		PrecMin	18.9	0.1	0.7	0.4	0.4	1.0	0.8	0.4	0.9	0.6	0.8	1.7	1.1	(17)
(18)		PrecStd	4.3	0.7	0.5	0.6	0.6	0.8	1.0	1.0	0.9	0.9	0.8	0.8	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.1	31.1	41.7	54.6	73.0	86.2	87.6	81.0	70.7	51.0	33.2	29.5	(19)
MCWB			24.9	29.5	36.3	42.0	56.3	64.1	68.0	66.0	58.3	44.7	31.6	28.6	(20)	
2%		DB	20.9	26.7	37.2	49.4	64.0	82.5	84.2	76.6	64.3	45.7	30.5	26.8	(21)	
		MCWB	19.5	25.3	32.8	38.9	52.3	64.2	67.2	63.8	55.9	42.5	29.2	25.6	(22)	
5%		DB	16.8	20.7	34.6	44.1	57.6	79.2	81.0	72.9	59.2	42.7	28.3	24.3	(23)	
		MCWB	15.8	19.6	31.5	36.5	48.7	62.9	65.1	62.2	53.6	40.2	26.8	23.3	(24)	
10%		DB	12.8	16.3	32.0	40.7	52.2	75.7	78.1	69.2	56.0	39.7	25.4	19.8	(25)	
		MCWB	11.8	15.3	29.5	34.5	44.1	62.1	64.2	60.9	51.4	37.2	24.1	18.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.6	29.7	37.5	43.1	56.9	69.2	73.8	68.3	60.3	45.8	31.9	28.6	(27)
MCDWB			26.1	30.7	40.7	52.4	67.7	77.2	83.5	77.4	66.3	48.4	32.9	29.0	(28)	
2%		WB	19.6	25.3	33.9	40.0	53.1	67.0	69.9	65.9	57.4	42.9	29.5	25.7	(29)	
		MCDWB	20.9	27.0	36.1	47.4	62.9	76.9	80.1	72.7	62.5	45.5	30.4	26.6	(30)	
5%		WB	15.8	19.7	32.0	37.5	49.3	65.2	67.6	63.9	54.4	40.4	27.0	23.3	(31)	
		MCDWB	16.8	20.8	34.1	43.0	56.5	75.0	76.1	70.2	59.1	42.4	28.2	24.4	(32)	
10%		WB	12.0	15.4	29.6	35.5	45.5	63.4	65.9	62.0	51.9	37.6	24.0	19.0	(33)	
		MCDWB	13.0	16.4	32.1	39.5	51.7	73.0	74.1	67.4	55.3	39.4	25.5	19.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.6	14.6	16.4	15.8	14.3	17.5	16.3	14.8	11.2	8.7	13.9	12.9	(35)
MCDBR			12.3	14.0	15.5	20.2	22.0	23.9	21.9	21.0	16.9	10.0	13.6	10.6	(36)	
MCWBR			12.0	13.3	12.9	13.9	13.4	10.6	9.4	10.3	9.6	8.3	13.4	10.4	(37)	
5% WB		MCDWR	12.8	14.2	13.8	16.9	20.3	18.2	18.0	16.9	14.6	9.1	13.0	10.3	(38)	
		MCWBR	12.5	13.6	12.0	12.6	13.3	10.1	9.4	9.8	9.3	8.0	13.2	10.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.253	0.294	0.334	0.357	0.374	0.387	0.444	0.410	0.352	0.312	0.243	0.233	(40)	
taud		1.908	1.986	1.984	2.045	2.246	2.381	2.192	2.265	2.401	2.236	1.958	1.871	(41)		
Ebn at Noon		113	195	234	258	266	265	243	241	237	194	116	59	(42)		
Edh at Noon		13	25	36	42	38	34	40	34	25	19	11	7	(43)		
(44)	All-Sky Solar Radiation	RadAvg	90	357	816	1332	1573	1755	1581	1084	639	259	111	41	(44)	
(45)		RadStd	12	28	73	161	174	164	161	124	87	42	22	5	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	+5.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page